Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/02/2020
PI Name:	Buckland, Daniel M.D., Ph.D.		
Project Title:	Automated Vascular Access for Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dan.buckland@duke.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-510-1295
Organization Name:	Duke University		
PI Address 1:	Emergency Medicine and Mechanical Engineering		
PI Address 2:	1121 W Main St, Ste 100		
PI Web Page:			
City:	Durham	**State:**	NC
Zip Code:	27701-1779	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	07/23/2020	**End Date:**	07/23/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
Contact Email:	jay.lemery@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC20K1433		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE APRIL 2021: SEE BRIDGEMAN, LEILA, WHO IS PRINCIPAL INVESTIGATOR. Original Principal Investigator (PI) was Daniel Buckland, MD, PhD; Leila Bridgeman, PhD, worked on the project from the beginning and now is PI due to Dr. Buckland's appointment to Intergovernmental Personnel Agreement, per Exploration Medical Capabilities element of Human Research Program. An important step in medical care and research of astronauts involves testing blood or giving intravenous medications. Placing a catheter into a blood vessel takes some degree of training and takes up crewmember's time. The goal of this project is to develop an automated technology that can identify blood vessels (such as a vein or artery), place a needle into the blood vessel, and then place a catheter into the blood vessel so that blood samples can be removed, and medications administered. The technology will operate under human supervision, but not require any active human control, to reduce the amount of training needed to operate it. The project will use a robotic arm with an attached ultrasound probe to automatically identify blood vessels using computer vision techniques, and then use that same robotic arm to insert the needle and catheter. Initial work will be done in simulations of human arms and then progress to human testing when determined safe. The significance of this work could lead to improved medical care of astronauts, as it will no longer require as many crew members or resources to start medical diagnosis and treatment. It could also allow for more complete automation or remote operation of medical care for astronauts. This technology could eventually help start medical care for patients on the ground who live in remote locations or do not have enough medical staff available.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This technology could eventually help start medical care for patients on the ground who live in remote locations or do not have enough medical staff available.
Task Progress:	New project for FY2020 (in July 2020). NOTE 4/26/2021: Original Principal Investigator (PI) was Daniel Buckland, MD, PhD; Leila Bridgeman, PhD, worked on the project from the beginning and now is PI due to Dr. Buckland's appointment to Intergovernmental Personnel Agreement, per Exploration Medical Capabilities element of Human Research Program.
Bibliography Type:	Description: (Last Updated:)